DATA ANALYTICS
FINAL PROJECT

Delhi Air Pollution Analysis Using Python

Exploring pollution trends, station-level differences, and associations with weather

Proposed tools

Python • Pandas • NumPy • Matplotlib • Seaborn • Jupyter Notebook / Google Colab

Proposal purpose

To define the research questions, data workflow, planned visualizations, and expected deliverables before beginning the analysis.

Prepared as a project plan. All findings are to be calculated from the selected dataset; no results are assumed in advance.

1. Project Overview

This project will use Python to explore air-quality observations collected at monitoring stations across Delhi. The analysis will examine how particulate pollution varies over time and across locations, how different pollutants relate to one another, and whether available weather variables are associated with pollution levels.

The project is designed as an exploratory data analysis (EDA) portfolio: it will combine data cleaning, descriptive statistics, time-based analysis, visual storytelling, and evidence-based conclusions. It will not assume that any one source or weather condition causes a particular pollution pattern.

2. Problem Statement

Delhi's air pollution can vary by season, time of day, and monitoring location. A large collection of high-frequency observations can be difficult to interpret without a consistent cleaning and analysis workflow. This project aims to turn station-level air-quality CSV files into a clear, reproducible analysis that identifies major patterns, compares locations fairly, and communicates findings through visualizations.

3. Objectives

• Collect and document the source files and their coverage.

• Audit data quality: column types, missing values, duplicates, timestamp coverage, and sampling intervals.

• Clean and combine selected station files while preserving station identifiers and source traceability.

• Analyze monthly, daily, and hourly patterns in key pollutants such as PM2.5 and PM10.

• Compare pollution distributions across monitoring stations.

• Explore relationships among pollutants and available weather variables.

• Produce at least 10 meaningful visualizations and at least 5 evidence-based findings.

• Deliver a reproducible Jupyter Notebook / Google Colab notebook and a concise final report.

4. Dataset and Scope

Proposed source: OpenCity's Delhi hourly air-quality reports collection, which lists station CSV files labelled "15 minute AQI Data for 2024-25." The project will verify the actual timestamps, station coverage, units, and completeness before finalizing the analysis period.

Potential fields: timestamp, station name/ID, PM2.5, PM10, NO, NO2, NOx, NH3, SO2, CO, ozone, and available weather fields such as temperature (AT), relative humidity (RH), wind speed (WS), wind direction (WD), rainfall (RF), solar radiation (SR), and pressure (BP). Actual field availability may vary by station.

Proposed scope: begin with a manageable group of stations to validate the workflow, then expand to additional stations if data quality and processing time allow. Use the full available date range only after checking coverage; do not assume that "2024-25" means a complete calendar year. **Source link:** OpenCity — Delhi hourly air-quality reports

5. Proposed Methodology

Step 1 — Data acquisition and documentation

Download the station CSV files; keep untouched source copies; record source URL, download date, station name, and file name.

Step 2 — Initial data audit

Inspect row and column counts, timestamp range, data types, missingness, duplicate rows, station labels, and time intervals. Check whether columns and units are consistent across stations.

Step 3 — Data cleaning

Parse timestamps consistently (including time zones); convert pollutant/weather fields to numeric where appropriate; standardize column names and station labels; identify invalid or sentinel values; quantify missingness; remove duplicates only when justified.

Step 4 — Combine and engineer features

Append compatible station files into a common table; retain station identifiers; derive date, month, hour, weekday, and season fields. Aggregate 15-minute observations to hourly or daily summaries when appropriate.

Step 5 — Exploratory analysis

Calculate descriptive statistics and valid-observation counts; examine pollutant distributions, station differences, time patterns, correlations, and associations with weather.

Step 6 — Visualize and interpret

Build at least 10 charts with clear units, titles, labels, and explanatory Markdown. Explain what each chart shows and what it cannot establish.

Step 7 — Findings and limitations

Write at least five findings supported by actual values, describe missing-data or coverage limitations, and distinguish association from causation.

6. Planned Visualizations

The following charts are planned. Final selection may be adjusted after the data audit, depending on available fields and data quality.

No.	Visualization	Purpose
1	Monthly mean PM2.5 and PM10 line chart	Identify seasonal changes and high-pollution months.
2	Top 10 stations by mean PM2.5 bar chart	Compare station-level averages with valid-reading counts.
3	Weekday × hour heatmap	Find recurring daily and weekly patterns.
4	PM2.5 box plots by station	Compare medians, spread, and outliers.
5	Pollutant correlation heatmap	Explore which pollutant measurements move together.
6	Temperature vs PM2.5 scatter plot	Examine a potential association with temperature.
7	PM2.5 during rain vs no-rain periods	Compare distributions using a defined rainfall threshold.
8	Average hourly pollutant profiles	Show how pollutant levels change through a typical day.
9	Monthly pollution-category distribution	Compare categories over time, if a defensible classification is available.
10	Monthly PM2.5 trends for 4–6 stations	Compare seasonal patterns across selected locations.

7. Research Questions and Expected Insight Areas

The project will test the following questions. The items below are investigation areas, not predetermined findings.

1. Seasonal variation

Which months have the highest and lowest mean PM2.5/PM10? How large is the difference, and how consistent is it across stations?

2. Station-level differences

Which stations have the highest and lowest average concentrations, and do their distributions differ beyond the mean?

3. Daily/weekly patterns

Do average concentrations vary by hour or weekday? Are patterns stable across months or seasons?

4. Pollutant relationships

Which pollutant pairs show the strongest positive or negative correlations, and are those relationships consistent across stations?

5. Weather associations

How are PM2.5 and PM10 associated with temperature, humidity, wind speed, and rainfall where those variables are sufficiently complete?

8. How Findings Will Be Reported

Each final insight will include (a) a direct statement, (b) a measured value or comparison, (c) the time period and stations covered, (d) a supporting chart or table, and (e) a short limitation or caveat where needed.

Example format (placeholders only): “The monthly mean PM2.5 peaked in [month] at [value] µg/m³, compared with [value] µg/m³ in [lowest month]. This is a [percentage]% difference based on [n] valid observations.” Values will be filled in only after analysis.

Important interpretation safeguards

- Do not claim that one station is the most polluted overall without checking data coverage and valid sample counts.
- Do not infer causation from correlation or simple before/after comparisons.
- Do not call PM2.5 concentration categories “official AQI” unless the analysis calculates AQI using a documented,appropriate method.
- Report missing values and inconsistent station coverage; avoid silently imputing large gaps.
- Check timestamp time zones before aggregating by hour or merging external data.

9. Expected Deliverables

Deliverable	Description
Cleaned analysis dataset	Combined table for selected stations with standardized columns and documented cleaning decisions.
Jupyter Notebook / Colab	Executable code, Markdown explanations, data audit, cleaning, analysis, charts, and findings.
10+ visualizations	Readable charts with labels, units, captions, and interpretation.
5+ evidence-based insights	Concise findings supported by calculations and visual evidence.
Project report / presentation	Summary of the problem, dataset, method, results, limitations, and conclusion.

Deliverable	Description
Source documentation	Dataset link, original file list, download date, and data dictionary/field notes where available.

10. Proposed Work Plan

Phase	Planned work	Output
1	Dataset download and audit	Coverage and quality summary
2	Cleaning and standardization	Documented clean dataset
3	Feature engineering and descriptive statistics	Analysis-ready table and summary tables
4	Build and refine visualizations	10+ charts with explanations
5	Interpret results and validate findings	5+ evidence-based findings
6	Finalize notebook and report	Submission-ready portfolio

11. Tools and Libraries

Python for analysis; **Pandas** and **NumPy** for data preparation and calculations; **Matplotlib** and **Seaborn** for charts; **Jupyter Notebook or Google Colab** for reproducible work. Additional libraries will be used only if required.

12. Proposed Project Outcome

The final outcome will be a reproducible Python-based analysis that documents data quality, explains temporal and station-level patterns, explores pollutant and weather relationships, and communicates at least five findings through at least ten visualizations. The final scope and conclusions will be based on the data audit and measured results.

Reference dataset collection

https://data.opencity.in/dataset/delhi-hourly-air-quality-reports

Reference format supplied for inspiration

Hidden Groundwater Contamination Patterns in Tamil Nadu Using Python (PDF)